

Jeremy Siu

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SKILLS

Languages: C/C++, Javascript, Python, HTML, CSS/SCSS, Bash, SQL

Frameworks/Libraries: Pytorch, scikit-learn, LangChain, Pandas, NumPy, React, Node.js

Technologies: Linux, Git, VS Code, Cursor, Google Cloud, AWS, PostgreSQL, Supabase, Jupyter, MATLAB

EXPERIENCE

Applied Machine Learning Engineer

Jan 2026 – Apr 2026

Government of Canada (Environment and Climate Change Canada)

Toronto, ON

- Developed 3+ hail forecasting models by designing ConvLSTM, U-Net, and Transformer deep learning architecture to improve baseline model accuracy by 33.7% and improve ROC-AUC by over 20%
- Engineered a GCP analytics pipeline for Random Forest and XGBoost cloud classifiers, increasing efficiency by 97% and saving \$6000+ annually
- Reduced classifier size by 190× from 3.8GB to 20MB by replacing Random Forest with XGBoost for portable deployment
- Implemented Optical Flow methods to validate cloud convection forecasting, reducing \$4000+ in verification costs
- Saved an estimated \$27,684 in staffing costs by independently delivering 3 ML projects beyond the original project scope

Front End Engineer & UX Designer

May 2025 – Aug 2025

Wiz Robotics

Richmond Hill, ON

- Built React frameworks with reusable component libraries to create Escape Room and Flash Card content 3× faster
- Refactored 50+ CSS files into a modular SCSS structure, eliminating cascade conflicts across 30+ components, improving maintainability and reducing front-end technical debt by 20%
- Designed and implemented an end-to-end testing workflow using Playwright, standardizing test creation for the development team, mitigating 100+ edge case errors across interactive React components
- Converted React projects into teachable curriculum, securing adoption by FIRST Robotics and reaching 500+ educators

Python and Arduino Instructor

May 2025 – Dec 2025

Wiz Robotics

Richmond Hill, ON

- Mentored 50+ students in Python with hands-on projects and customized lessons to improve technical skills and confidence
- Guided students in completing embedded software final projects, integrating Arduino, sensors, and hardware control systems

PROJECTS

Forecast: Data Automation Platform (Winner @FCI Hackathon) | *LangChain, Playwright, FastAPI*

[Demo Link](#)

- Designed an AI centric pipeline to extract data for Waterloo's readiness for a population of 1 million by developing a RAG pipeline to calculate a composite score against 50+ benchmarks and relations to each of the 5 categories
- Automated data collection with Playwright and recorded web scraping sessions for data validation and user transparency
- Awarded \$2000 for Best Technical use of LangChain among 46 projects and 128 participants

Callio Labs: Agentic Genome Research (Winner @GenAI Genesis) | *LangFlow, FastMCP, ColabFold*

[Demo Link](#)

- Developed an agentic AI pipeline equipped with MCP tools calls, NCBI Datasets, and Primer3 APIs to run 5+ parallelized tests to determine optimal primer sequences, significantly reducing research time
- Awarded the Best Health Care Hack using Agentic AI and Top 5 Project among 249 projects and over 800 participants

OrsusHealth: Heart Disease Prediction App | *Scikit-learn, AWS, PostgreSQL, React*

[Demo Link](#)

- Trained multiple probabilistic machine learning models including Logistic Regression and Random Forest models for heart disease prediction to improve ROC-AUC score to 0.95 and accuracy to 89%
- Integrated AWS Lambda to host the model and provide real-time predictions with Fast API in under 10 seconds
- Designed an AI agent using LangChain to equip the model with a clinical facts database and patient profiles by integrating MCP tool calls to provide realistic, user-centric data to improve heart health

Compass: See Insights Clearer (Winner @QHacks) | *PostgreSQL, Gadium API, Rive*

[Demo Link](#)

- Engineered a custom RAG pipeline using pgvector, Gemini, and sentence-transformers to identify hidden relations between 137,000+ service requests to simultaneously solve pressing issues and reduce data processing time
- Integrated Gadium and Gemini API for animated text-to-speech for easy interaction with real-time AI analysis
- Deployed a cloud-hosted FastAPI backend on Vultr to automate daily batch retrieval of 1,000,000+ records with cron jobs

Avesia: Making Cameras Smart (Winner @NexHacks) | *Overshoot SDK, React Flow, Gemini API*

- Built a no-code AI surveillance workflow builder for creating custom CV detection pipelines, event triggers, and automated alert flows winning 1st place among 874 participants and 100+ projects

EDUCATION

University of Waterloo

GPA: 4.0

Bachelor's of Biomedical Engineering

Relevant Coursework: Data Structures and Algorithms, Human Factors Engineering, Digital Computation